FINANCIAL MANAGEMENT
Suggested Answers
March-April 2023

Answer to the Question# 1(a) (i)

Market capitalization
Workings
Return on equity per equity
$r = rf + \beta(rm - rf)$;
$rm - rf$ = 2.5%; β = 1.2; $and\ rf$ = 8%
r = 8+ (1.2 × 2.5) = 11%

Dividend paid is BDT 5 per share

P = (Dividend paid/ return on equity) = 5/ .11 = BDT 45.45 = 45.5 per share

Capitalization		**BDT '000**
Equity Shares (10,000,000)	(10,000,000x45.5)	455,000
14% Preference shares		250,000
12% Bonds – Irredeemable	(920 / 1,000 × 200,000)	184,000
Total Capitalization		**889,000**

Answer to the Question# 1(a) (ii)

WACC before investment = 11.59%

	Capital '000	**Cost**	**Amount'000**
Equity Shares	455,000	11.00%	50,050
14% Preference shares	250,000	14.00%	35,000
12% Bonds – Irredeemable	184,000	9.78%	18,000
Total	**889,000**	**11.59%**	**103,050**

Cost of bonds is (12% * 1000)/920 X (1 – 25%) = 9.78%

WACC after investment= 11.39%

Bond amount to be raised = 100,000,000 X (1,000/920) **= BDT 108,695,652**

	Capital '000	**Cost**	**Amount'000**
Equity Shares	455,000	11.00%	50,050
14% Preference shares	250,000	14.00%	35,000
12% Bonds – Irredeemable	292,696	9.78%	28,633
Total	**997,696**	**11.39%**	**113,683**

Answer to the Question# 1(a) (iii)

Share Buy-back:

It is the process where a company buys back its own share from investors or from the stock market at market prevailing rate.
Companies may purchase their own shares for the following reasons

Discourage unfriendly takeover
Shares repurchased can be used to enhance shareholder value or discourage an unfriendly takeover.

Acquire another business
Companies can repurchase their own shares to be used for acquisition of another company.

For employees share options
Where shares are used as part of employee salary package, a company can purchase its own shares for that purpose.

Redistribution of excess cash
Companies with excess cash as a result of excessive retained earnings can redistribute the excess cash to shareholders through purchase of own shares.

Increase shareholder value
By reducing the number of shares outstanding and thus increasing earnings per share and return on equity

As part of agreement with investors (Share repurchase agreement)

Where there is an agreement with an investor that the company will buy its shares from the investor after a given period, the company will have to comply with this agreement by purchasing its own shares from the investor. This is also referred to as share repurchase agreement.

Answer to the Question# 1(a) (iv)

Profitable for the Company?

iv) The decision to buy back shares should be based on several factors, such as the company's financial position, cash flow, and profitability. In the case of Karnofuli Ltd., the company has a low gearing level and a cost of equity that is higher than the cost of debt, which suggests that it may be beneficial for the company to increase its leverage. However, the decision to buy back shares should also take into account the company's future investment opportunities and the potential impact on dividend payments. If the company has good investment opportunities and needs to retain cash for future growth, then share buy-back may not be the best option. Additionally, the company's ability to generate cash flow should also be considered, as buying back shares may strain the company's cash reserves. Therefore, before deciding to buy back shares, the company should conduct a thorough analysis of its financial position and future prospects to determine if it is a good choice.

Answer to the Question# 1 (b) i, ii & iii

Accounts receivable changes and bad debts.

i)	Bad debts	
	Proposed plan (1,137,600 × 0.04)	45,504
	Present plan (948,000 × 0.02)	18,960
ii)	Incremental Bad debts	26,544

iii) Yes, because the additional bad debts and interest costs on additional credit sales are lower than net benefits from the incremental sales as calculated below:

Additional profit/contribution from sales:			
9480 additional units (Tk. 20 – Tk.15)			47,400
Incremental Bad debts (from part ii)			26,544
			20,856
Interest cost on additional credit sales/ period	(Tk. 1137600/365 X 0.12 X 66 days) -	24,684	5,984
	(Tk. 948000/365 X 0.12 X 60 days)	(18,700)	
Net benefit from implementing proposed plan			14,872

Workings:

Current Credit Sales is Taka	1,580,000 X 60%	= Taka 948,000
New Credit Sales is Taka	948,000 X 1.20	= Taka 11,37,600
Incremental Credit Sales	(11,37,600 – 948,000) or Taka 189,600 or 9480 units	

Recommendation:
When additional sales are ignored, the proposed policy is rejected. However, when all the benefits are included, the profitability from new sales outweigh the increased interest (opportunity) cost and bad debts. Therefore, the policy is recommended.

Alternatively:

	Existing	**Proposed**	**Incremental**
Total Sales in Taka	1,580,000		
Unit Sales price in Taka	20		
Credit Sales in Taka	948,000	1,137,600	
Credit Sales in Unit	47,400	56,880	
Average collection period (days)	60	66	
Bad debts % of credit sales	2	4	
Bad debts in Taka	18,960	45,504	26,544
Incremental interest cost on credit sales & for increased collection period in Taka	18,700 [(948000/365)*60*12%]	24,684 [(1137600/365)*66*12%]	5,984

Incremental profit:

Contribution in Taka [9480 X (20-15)]	47,400
Incremental Bad Debts in Taka	(26,544)
Incremental Interest (opportunity) cost in Taka	(5,984)
Net Incremental Profit in Taka	14,872

Recommendation:
Since there will be net profit after Bad Debts and Interest (opportunity) cost, therefore the proposed plan is recommended.

i) The Modigliani and Miller (MM) view is that dividend policy is irrelevant to the firm's cost of capital and stock prices. They argue that the total value of a firm is determined by its earnings power and risk, and not by the way it distributes its earnings between dividends and retained earnings. On the other hand, Gordon and Lintner (GL) believe that dividend policy affects both the cost of capital and stock prices. They argue that investors prefer a stable and predictable

stream of dividends, and firms that pay high dividends are perceived as less risky and hence have a lower cost of capital.

ii) The tax preference theory differs from the views of MM and GL in that it suggests that investors prefer capital gains to dividends because capital gains are taxed at a lower rate than dividends. This theory implies that firms should retain earnings and reinvest them rather than pay dividends.

iii) MM could use the information content or signaling hypothesis to argue that dividend policy does matter because it signals important information to investors. For example, a firm that increases its dividend sends a positive signal to investors that it is confident in its future prospects. If I were debating MM, I would counter by arguing that investors are rational and can obtain information about a firm's prospects through other means, such as financial statements and analyst reports.

iv) MM could use the clientele effect concept to argue that firms should not change their dividend policy because it may upset the existing clientele of investors who prefer a certain level of dividends. For example, if a firm reduces its dividend, it may lose some of its existing investors who rely on that dividend income. If I were debating MM, I would counter by arguing that the clientele effect is not significant in today's market because investors have access to a wide range of investment options and can easily switch between different stocks and securities.

Answer to the Question# 2(a) (i)

Profit/Loss of exercising: (OPTION A)

Cost of exercising

		BDT
$75m @ 107	=	8,025,000,000
Premium paid	=	2,250,000
TOTAL		**8,027,250,000**

Cost of not Exercising and buying from the spot market

		BDT
$75m @ 105	=	7,875,000,000

Padma Limited should not exercise the option; but buy from the spot market That will be cheaper not withstanding option premium already paid. Total loss from this option (by not exercising) will be BDT 2,250,000. Exercising Option A would cost 150 million (without premium).

Profit/Loss of exercising: (OPTION b) Cost of exercising

Cost of exercising

		BDT
Eur 65 m @ 110	=	7,150,000,000
Premium paid	=	2,275,000
TOTAL		**7,152,275,000**

Cost of not Exercising and buying from the spot market

		BDT
Eur 65m @ 112	=	7,280,000,000

Gain from option B by exercising 127,725,000
Option B should be exercised.

Answer to the Question# 2(a) (ii)

Overall gain/ loss:		
Option A	Loss	2,250,00
Option B	Gain	**127,725,000**
Total	Gain	**125,475,000**

Answer to the Question# 2(b) (i)

Report on Managing Foreign Exchange Risk for Bangla Metal Ltd.

Introduction: Bangla Metal Ltd. has recently been awarded a large contract with a European Company, which will be paid in Euros and with a credit period of 3 months. As an accountant of the company, I am concerned about the foreign exchange risk associated with this transaction. This report aims to highlight the points that need to be considered in relation to foreign exchange risk and the available solutions to manage such risks.

Foreign Exchange Risk: Foreign exchange risk refers to the potential financial loss that a company may suffer due to changes in foreign currency exchange rates. Such risks can be caused by various factors such as economic conditions, political events, and market fluctuations. In this case, Bangla Metal Ltd. is exposed to foreign exchange risk due to the volatile nature of the Euro and BDT.

Available Solutions: To manage the foreign exchange risk, Bangla Metal Ltd. can consider the following options:

i) Do nothing and accept the risk: One option for the company is to do nothing and accept the foreign exchange risk associated with this transaction. This means that the company will be exposed to the risk of losing a substantial amount due to exchange rate fluctuations. Doing nothing may seem like a simple and costless option, but the potential financial loss can be significant.

ii) Using a forward exchange contract: Another option for the company is to use a forward exchange contract. This is a contract between the company and a bank or financial institution to exchange a fixed amount of currency at a predetermined exchange rate on a future date. This can help the company to lock in the exchange rate and eliminate the risk of exchange rate fluctuations. However, the cost associated with a forward exchange contract is the difference between the spot rate and the forward rate.

iii) Carrying out a market hedge: A market hedge involves taking an offsetting position in the foreign exchange market to protect against foreign exchange risk. This can be done by buying or selling currencies in the spot or futures market. The cost associated with a market hedge is the transaction cost, which is the fee charged by the bank or financial institution for carrying out the hedge.

iv) Buying a currency option: The company can also consider buying a currency option. This is a contract that gives the company the right, but not the obligation, to buy or sell a fixed amount of currency at a predetermined exchange rate on a future date. This can help the company to protect against exchange rate fluctuations while allowing the company to benefit from favorable exchange rate movements. The cost associated with a currency option is the premium paid to the seller of the option.

Conclusion: In conclusion, foreign exchange risk is a potential financial loss that Bangla Metal Ltd. needs to consider when dealing with the European Company. The available solutions to manage such risks include doing nothing, using a forward exchange contract, carrying out a market hedge, and buying a currency option. Each option has associated costs that the company needs to consider before deciding on the appropriate solution. It is recommended that the company seeks professional advice before making any decisions.

Answer to the Question# 3(a)

Accounting profit

	Liza Textile	**Shormi Textile**
Accounting profit	BDT	BDT
Cash inflow	13,000,000	32,000,000
Depreciation	(5,291,667)	(12,166,667)
Annual profit	**7,708,333**	**19,833,333**
Avergage Book Value	17,125,000	38,500,000
ARR		
Using average value	45%	52%

Decision:
Shormi Textile gives the highest Rate of return and should be selected

Answer to the Question# 3(b)

Net Present Value (NPV) method NPV @ 9%

	Liza Textile	**Shormi Textile**
	BDT	BDT
Year 0 Cost	(33,000,000)	(75,000,000)
Year 1-6 cash flow @ annuity factor of 4.486	58,318,000	143,552,000
Year 6 scrap value @ 0.596	745,000	1,192,000
	26,063,000	**69,744,000**

Both projects produced positive NPVs but Shormi textile has the highest NPV and should be selected since both projects are mutually exclusive.

Answer to the Question# 3(c)

Net Present Value (NPV) method NPV @ 40%

	Liza Textile	**Shormi Textile**
	BDT	BDT
Year 0 Cost	(33,000,000)	(75,000,000)
Year 1-6 cash flow	28,184,000	69,376,000
Year 6 scrap value	166,250	266,000
	(4,649,750)	**(5,358,000)**
IRR	**35.31%**	**37.79%**

Both projects give IRR above the 9% cost of Capital, Shormi Textile gives the highest 38% and should be taken since the two projects are mutually exclusive

Answer to the Question# 3(d)

Payback period	**Liza Textile**	**Shormi Textile**
Cost of machine	33,000,000	75,000,000
Cash inflow	13,000,000	32,000,000
Payback period	**2.54**	**2.34**

Shormi textile gives the lowest payback period and should be selected.

Answer to the Question# 4(a)

NPV Calculation

	0	1	2	3	4	5
Investment	(115,000,000)					
Contract			55,000,000	55,000,000	55,000,000	55,000,000
Sales proceeds		12,000,000				
Savings			2,500,000	2,500,000	2,500,000	2,500,000
Adhoc revenue			2,700,000	4,200,000	5,700,000	7,200,000
Sale proceeds						12,000,000
operating costs			(20,195,000)	(20,720,000)	(21,245,000)	(21,770,000)
Pre-tax cash	**(115,000,000)**	**12,000,000**	**40,005,000**	**40,980,000**	**41,955,000**	**54,930,000**
Tax			(10,001,250)	(10,245,000)	(10,488,750)	(13,732,500)
Tax saved on TDA	5,750,000	4,600,000	3,680,000	2,944,000	2,355,200	6,420,800
Net cash flow	**(109,250,000)**	**16,600,000**	**33,683,750**	**33,679,000**	**33,821,450**	**47,618,300**
Discount factor	1.000	0.926	0.857	0.794	0.735	0.681
PV	(109,250,000)	15,371,600	28,866,974	26,741,126	24,858,766	32,428,062
NPV	**19,016,528**					

Answer to the Question# 4(b)

Additional factors

The appropriateness of the discount factor and the impact of this new venture on risk profile (financial and/or business).

The accuracy of the estimates employed across the board and the potential impact of inflation.

Whether, given the commercial risks involved, the company really needs to do this work itself.

The potential impact of exchange rate fluctuations on the project: this suggests that the whole issue of managing such risks needs to be addressed by the directors, and the cost of any such actions needs to be considered.

The directors must note that a large proportion of the positive NPV is dependent on the final sale proceeds.

The advice to the directors should be to proceed with the investment in view of the positive NPV, subject to satisfaction with regard to the issues raised above.

Answer to the Question# 5(a)

The company is currently an all-equity firm, so the value as an all-equity firm equals the present value of after-tax cash flows, discounted at the cost of the firm's unlevered cost of equity. So, the current value of the company is:

V_U = [Pretax earnings) (1 – tc)] / R_0

V_U = [(Tk. 21,000,000) (1 - .35) / .16

V_U = Tk. 85,312,500

The price per share is the total value of the company divided by the shares outstanding, or:

Price per share = Tk. 85,312,500 / 1,300,000

Price per share = Tk. 65.63

Answer to the Question# 5(b)

The adjusted present value of a firm equals its value under all-equity financing plus the net present value of any financing side effects. In this case, the NPV of financing side effects equals the aftertax present value of cash flows resulting from the firm's debt. Given a known level of debt, debt cash flows can be discounted at the pretax cost of debt, so the NPV of the financing effects are:

NPV = Proceeds – Aftertax PV (Interest Payments)

NPV = Tk. 30,000,000 – (1 – 35) (.09) (Tk. 30,000,000) / .09

NPV = Tk. 10,500,000

So, the value of the company after the recapitalization using the APV approach is:

V = Tk. 85,312,500 + 10,500,000

V = Tk. 95,812,500

Since the company has not yet issued the debt, this is also the value of equity after the announcement. So, the new price per share will be:

New share price = Tk. 95,812,500 / 1,300,000

New share price = Tk. 73,70

Answer to the Question# 5(c)

The company will use the entire proceeds to repurchase equity. Using the share price we calculated in part b, the number of shares reprehend will be:

Shares repurchased = Tk. 30,000,000 / Tk. 73.70

Shares repurchased = 407,045

And the new number of shares outstanding will be:

New shares outstanding = 1,300,000 – 407,045

New shares outstanding = 892,955

The value of the company increased, but part of that increase will be funded by the new debt. The value of equity after recapitalization is the total value or the company minus the value of debt, or:

New value of equity = Tk. 95,812,500 = 30,000,000

New value of equity = Tk. 65,812,500

So, the price per share of the company after recapitalization will be:

New value of equity = Tk. 65,812,500 / 892,955

New value of equity = Tk. 73.70

The price per share is unchanged.

---The End---